

AP Statistics MCQ Practice Exam 04

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AP Statistics MCQ Practice Exam 4

PART A, QUESTIONS

APSTAT-MCQ4-001

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A two-proportion z-test compares compliance rates for training method A and training method B using the statistic based on $\hat{p}_A - \hat{p}_B$. The result is $z = -2.10$ with two-sided p-value 0.0357. If the groups are reversed so the statistic is based on $\hat{p}_B - \hat{p}_A$, what are the new z statistic and p-value?

- A) $z = -2.10$ and p-value 0.9643
- B) $z = 2.10$ and p-value 0.0357
- C) $z = 2.10$ and p-value 0.0179
- D) $z = -0.476$ and p-value 0.0357

APSTAT-MCQ4-002

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A school district deliberately samples equal numbers of families from four home-language groups, even though the groups make up different percentages of district enrollment. To estimate an overall district proportion, what should analysts do?

- A) Treat every sampled family equally because each language group has the same sample size.
- B) Discard responses from the smallest language groups until sample shares match enrollment shares.
- C) Report only four separate group estimates because no overall estimate is possible.
- D) Weight group results according to the groups' shares of district enrollment.

APSTAT-MCQ4-003

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For 25 employees, the difference in task time is defined as new method – old method. The sample mean difference is –3.2 minutes and the sample standard deviation of the differences is 4.5 minutes. For a 90% paired t-confidence interval with $df = 24$, use $t^* = 1.711$. Which interval estimates the population mean difference?

- A) (–4.74, –1.66) minutes
- B) (–4.00, –2.40) minutes
- C) (–7.70, 1.30) minutes
- D) (–10.90, 4.50) minutes

APSTAT-MCQ4-004

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

A transit planner simulates 10,000 days under a proposed bus schedule. In 2,740 simulated days, at least one rider misses a connection. Which interpretation is appropriate?

- A) Exactly 2,740 of every future 10,000 days will have a missed connection.
- B) The true probability of at least one missed connection is known to equal 0.274.
- C) The simulation estimates the probability of at least one missed connection as 0.274, with some Monte Carlo variability.
- D) The probability that the simulation method is correct is 0.274.

APSTAT-MCQ4-005

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For adults ages 18 through 80, the correlation between age and a reaction-speed score is $r = -0.70$. Among only adults ages 35 through 45, the correlation is $r = -0.24$. Which explanation is most plausible?

- A) Restricting the range of age can weaken the observed correlation even if the broader linear tendency still exists.
- B) Correlation must become positive whenever the range of x is reduced.

- C) The smaller correlation proves that age has no association with reaction speed for any adult.
- D) The two correlations cannot differ because correlation is unchanged by selecting a subgroup.

APSTAT-MCQ4-006

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A help desk found that 84 of 120 service requests submitted through its mobile app and 90 of 150 requests submitted through its desktop site were resolved during the first contact. Which comparison is correct?

- A) Desktop has a 6-percentage-point higher first-contact resolution rate.
- B) Desktop has a 10-percentage-point higher first-contact resolution rate.
- C) Mobile has a 10-percentage-point higher first-contact resolution rate.
- D) Mobile has a 24-percentage-point higher first-contact resolution rate.

APSTAT-MCQ4-007

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

Let p_E be the population proportion of participants who complete a certification course with an email coach and p_U the proportion who complete it without a coach. Which research question corresponds to $H_a: p_E - p_U > 0$?

- A) Is the completion proportion the same in the two populations?
- B) Is either population completion proportion different from 0.50?
- C) Is the uncoached completion proportion higher than the coached proportion?
- D) Is the coached completion proportion higher than the uncoached proportion?

APSTAT-MCQ4-008

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ4-008 through APSTAT-MCQ4-010 use the following stimulus. For 28 archaeological survey plots, x is soil depth in centimeters and y is artifact density per square meter. The scatterplot has a moderately strong negative linear pattern with $r = -0.74$. The following candidates include the least-squares regression line and have these sums of squared vertical residuals:

Candidate P: $SSE = 410$

Candidate Q: $SSE = 285$

Candidate R: $SSE = 330$

Which candidate is the least-squares regression line?

- A) Candidate P, because its SSE is largest
- B) Candidate Q, because its SSE is smallest
- C) Candidate R, because its SSE is between the other values
- D) All three, because correlation is the same for the dataset

APSTAT-MCQ4-009

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ4-008 through APSTAT-MCQ4-010 use the following stimulus. For 28 archaeological survey plots, x is soil depth in centimeters and y is artifact density per square meter. The scatterplot has a moderately strong negative linear pattern with $r = -0.74$. The following candidates include the least-squares regression line and have these sums of squared vertical residuals:

Candidate P: $SSE = 410$

Candidate Q: $SSE = 285$

Candidate R: $SSE = 330$

Why are the residuals described as vertical when predicting artifact density from soil depth?

- A) Each residual is observed y – predicted y at the same x -value, so it is a vertical difference on the scatterplot.
- B) Each residual is observed x – predicted x at the same y -value, so it is a horizontal difference.
- C) Residuals are the perpendicular distances from points to a line.
- D) Residuals measure only the distance between each x -value and $\bar{x}$.

APSTAT-MCQ4-010

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ4-008 through APSTAT-MCQ4-010 use the following stimulus. For 28 archaeological survey plots, x is soil depth in centimeters and y is artifact density per square meter. The scatterplot has a moderately strong negative linear pattern with $r = -0.74$. The following candidates include the least-squares regression line and have these sums of squared vertical residuals:

Candidate P: $SSE = 410$

Candidate Q: $SSE = 285$

Candidate R: $SSE = 330$

If the roles of soil depth and artifact density are reversed, which statement is correct?

- A) The correlation changes to 0.74, and the new regression line is the algebraic inverse of the old line.
- B) The correlation changes to $-1/0.74$, and the original regression line is unchanged.
- C) The correlation becomes zero because the response variable changed.
- D) The correlation remains -0.74 , but the new least-squares line is not generally the algebraic inverse of the original line.

APSTAT-MCQ4-011

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A project team has 7 remote members and 5 office-based members. Two different members are selected at random without replacement. What is the probability that both selected members are remote?

- A) $7/24$
- B) $7/44$
- C) $7/12 \times 6/11 = 7/22$
- D) $(7/12)^2 = 49/144$

APSTAT-MCQ4-012

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

The thicknesses of ice samples have mean 18.0 millimeters and standard deviation 4.2 millimeters. Which statement best interprets the standard deviation?

- A) Every sample thickness is within 4.2 millimeters of 18.0 millimeters.
- B) Sample thicknesses typically differ from the mean thickness by about 4.2 millimeters.
- C) Exactly 68% of sample thicknesses are between 13.8 and 22.2 millimeters.
- D) The middle 50% of sample thicknesses spans 4.2 millimeters.

APSTAT-MCQ4-013

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

An auditor takes a simple random sample of 80 account balances without replacement from a branch containing 500 accounts and plans a one-sample t-procedure for the population mean balance. What is the main concern?

- A) A t-procedure cannot be used for monetary data.
- B) The sample must contain exactly half the branch accounts.
- C) The population standard deviation must be known before sampling.
- D) Because 80 exceeds 10% of 500, dependence from sampling without replacement may make the usual standard-error condition questionable.

APSTAT-MCQ4-014

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

Why does a two-proportion z-test of $H_0: p_1 = p_2$ use a pooled sample proportion in its standard error, while a confidence interval for $p_1 - p_2$ generally does not?

- A) Under H_0 , both samples are assumed to share one common population proportion, which the pooled proportion estimates.
- B) Pooling is required whenever the two sample sizes are unequal.
- C) A confidence interval assumes both population proportions equal zero.
- D) Pooling makes the observed difference $\hat{p}_1 - \hat{p}_2$ equal to zero.

APSTAT-MCQ4-015

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

A bakery's daily revenue has population mean μ and standard deviation σ . A manager will select 64 days at random and wants the probability that their average revenue exceeds a target. Which random variable and sampling distribution should be used?

- A) The revenue from one randomly selected day, centered at $\mu/64$
- B) The total revenue from 64 days, with standard deviation $\sigma/\sqrt{64}$
- C) The sample mean $\bar{x}$, centered at μ with standard deviation $\sigma/\sqrt{64}$
- D) The sample proportion $\hat{p}$, centered at μ with standard deviation $\sigma/64$

APSTAT-MCQ4-016

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Weekly household electricity use is approximately normal with mean 210 kilowatt-hours and standard deviation 30 kilowatt-hours. What proportion of households use less than 165 kilowatt-hours in a week? Use $\Phi(-1.50) = 0.0668$.

- A) 0.9332
- B) 0.0668
- C) 0.1587
- D) 0.5000

APSTAT-MCQ4-017

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

Two measurements taken on the same subjects have a strong positive association. A researcher compares the conditions using paired differences rather than incorrectly treating the measurements as independent samples. What is a likely benefit of pairing?

- A) Pairing changes a quantitative response into a categorical response.
- B) Pairing guarantees that the population mean difference is zero.
- C) Subject-to-subject variability can cancel in the differences, often reducing the standard error.

D) Pairing makes random assignment unnecessary in every experiment.

APSTAT-MCQ4-018

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A workplace survey asks employees whether they completed required safety training. Managers will compare the survey responses with confidential training records for a random subset of respondents. What is the main statistical purpose of this record check?

- A) To increase the sample size without contacting more employees
- B) To turn the categorical response into a quantitative measurement
- C) To make the original respondents a simple random sample
- D) To assess possible response or classification error in the self-reported training status

APSTAT-MCQ4-019

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Formulate Questions

Prompt:

An engineer wants a regression model that will predict sound absorption from material density for products with densities from 20 to 80 kilograms per cubic meter. Which data-collection plan best supports that goal?

- A) Measure both variables on products sampled across the full intended density range, with adequate replication throughout that range.
- B) Measure sound absorption repeatedly on one product with density 50 kilograms per cubic meter.
- C) Select only products with the highest observed sound absorption.
- D) Record density for many products but estimate sound absorption from their brand names.

APSTAT-MCQ4-020

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

To study financial habits in a population that is difficult to identify from public records, a researcher begins with several acquaintances and asks each participant to recruit others from the same social network. What is the chief limitation of this snowball sample?

- A) It always produces a sample larger than the population.
- B) People connected to the initial participants may be overrepresented, so selection probabilities are unknown.
- C) It is equivalent to a stratified random sample by social network.
- D) It guarantees nonresponse bias but eliminates selection bias.

APSTAT-MCQ4-021

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Two irrigation valves operate independently. Valve A opens successfully with probability 0.90, and valve B opens successfully with probability 0.80. What is the probability that exactly one valve opens successfully?

- A) 0.02
- B) 0.18
- C) 0.72
- D) 0.26

APSTAT-MCQ4-022

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

For a fixed sample size and fixed true alternative proportion, what generally happens when the significance level is lowered from $\alpha = 0.10$ to $\alpha = 0.01$?

- A) Both Type I and Type II error probabilities decrease.
- B) Type I error probability increases, and power increases.
- C) Type I error probability decreases, but Type II error probability generally increases.
- D) Type I error probability remains fixed because the sample size is unchanged.

APSTAT-MCQ4-023

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A 95% one-sample t-confidence interval for a population mean is (14.2, 18.6). The critical value used was $t^* = 2.064$. What standard error was used to construct the interval?

- A) 1.07
- B) 2.20
- C) 4.40
- D) 8.80

APSTAT-MCQ4-024

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A histogram summarizes 40 customer order values using the following bins and frequencies:

- \$0–9: 4 orders
- \$10–19: 9 orders
- \$20–29: 15 orders
- \$30–39: 8 orders
- \$40–49: 4 orders

In which interval must the median order value lie?

- A) \$0–9
- B) \$10–19
- C) \$30–39
- D) \$20–29

APSTAT-MCQ4-025

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ4-025 through APSTAT-MCQ4-027 use the following stimulus. An archive contains 12 film reels, 5 of which have nitrate damage and 7 of which do not. Four different reels are selected uniformly at random without replacement. Useful values are $C(12,4) = 495$, $C(5,2) = 10$, $C(7,2) = 21$, and $C(7,4) = 35$.

What is the probability that exactly 2 of the 4 selected reels have nitrate damage?

- A) $35/495 \approx 0.071$
- B) $210/495 \approx 0.424$
- C) $280/495 \approx 0.566$
- D) $420/495 \approx 0.848$

APSTAT-MCQ4-026

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ4-025 through APSTAT-MCQ4-027 use the following stimulus. An archive contains 12 film reels, 5 of which have nitrate damage and 7 of which do not. Four different reels are selected uniformly at random without replacement. Useful values are $C(12,4) = 495$, $C(5,2) = 10$, $C(7,2) = 21$, and $C(7,4) = 35$.

What is the probability that at least 1 of the 4 selected reels has nitrate damage?

- A) $35/495 \approx 0.071$
- B) $460/495 \approx 0.929$
- C) $245/495 \approx 0.495$
- D) $490/495 \approx 0.990$

APSTAT-MCQ4-027

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ4-025 through APSTAT-MCQ4-027 use the following stimulus. An archive contains 12 film reels, 5 of which have nitrate damage and 7 of which do not. Four different reels are selected uniformly at random without replacement. Useful values are $C(12,4) = 495$, $C(5,2) = 10$, $C(7,2) = 21$, and $C(7,4) = 35$.

Why are the damage statuses of the first and second selected reels not independent?

- A) The reels are selected uniformly, so later outcomes become deterministic.

- B) Independence is impossible whenever a sample contains more than one item.
- C) After the first reel is selected, the composition of the remaining reels changes, so the probability of damage on the second draw can change.
- D) The probability of damage on the second draw is always $\frac{5}{12}$ regardless of the first draw.

APSTAT-MCQ4-028

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

Two independent random samples are used to construct a confidence interval for $p_1 - p_2$. Sample 1 has 8 successes and 42 failures; sample 2 has 15 successes and 35 failures. Which condition is not satisfied?

- A) Sample 1 does not have at least 10 successes.
- B) Sample 1 does not have at least 10 failures.
- C) Sample 2 does not have at least 10 successes.
- D) Sample 2 does not have at least 10 failures.

APSTAT-MCQ4-029

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

A horticulturist wants to study the distribution of time to first bloom for a new flower variety. Which observational unit and variable are appropriate?

- A) Observational unit: day; variable: number of flowers in the entire greenhouse
- B) Observational unit: greenhouse; variable: whether any plant ever blooms
- C) Observational unit: flower variety; variable: number of horticulturists observing it
- D) Observational unit: individual plant; variable: days from sowing until its first bloom

APSTAT-MCQ4-030

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For a one-sample t-test of $H_0: \mu = 50$, a sample of $n = 16$ has $s = 8$ and test statistic $t = -2.00$. What is the sample mean $\bar{x}$?

- A) 42
- B) 48
- C) 46
- D) 54

APSTAT-MCQ4-031

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Across 60 neighborhoods, the correlation between the neighborhood mean number of books per household and the neighborhood mean literacy score is 0.82. Which conclusion is warranted?

- A) Every household with more books has a higher literacy score than every household with fewer books.
- B) Neighborhoods with higher mean book counts tend to have higher mean literacy scores, but the aggregate correlation does not determine the relationship for individual households.
- C) Increasing one household's book count will cause its literacy score to rise by 0.82 points.
- D) About 82% of households have literacy scores above their neighborhood mean.

APSTAT-MCQ4-032

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A youth program wants to determine whether mentoring raises the population completion proportion. Let p_M be the completion proportion under mentoring and p_C the completion proportion under the current program. Which formulation matches the question?

- A) Parameter $p_M - p_C$ with $H_a: p_M - p_C > 0$
- B) Parameter $\hat{p}_M - \hat{p}_C$ with $H_a: \hat{p}_M - \hat{p}_C = 0$
- C) Parameter $p_M + p_C$ with $H_a: p_M + p_C > 1$
- D) Parameter $p_C - p_M$ with $H_a: p_C - p_M > 0$

APSTAT-MCQ4-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A researcher randomly selects residential addresses for a mail survey. When a selected household does not respond, the researcher replaces it with the nearest responding household. What problem does this replacement create?

- A) It converts the response variable from categorical to quantitative.
- B) It guarantees that the final sample size will be too small.
- C) The replacements are not randomly selected and may systematically differ from nonresponding households.
- D) It produces a census rather than a sample.

APSTAT-MCQ4-034

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

A warehouse has 50 specialized parts, 12 of which are refurbished. A manager wants to use a binomial model to simulate the number of refurbished parts in 20 selections. Which revision would best satisfy the independence and constant-probability conditions?

- A) Select 20 parts without replacement and ignore the changing composition.
- B) Select all 50 parts and stop after observing 20 refurbished parts.
- C) Arrange the parts by age and select every second part.
- D) Select one part at random, record its status, replace and remix it, and repeat 20 times.

APSTAT-MCQ4-035

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Independent random samples of sizes 60 and 55 are taken from two populations. Both sample distributions are moderately right-skewed with no outliers. Which statement best supports using a two-sample t-procedure?

- A) The large sample sizes make the procedure reasonably robust to the moderate skewness.
- B) A t-procedure requires both sample distributions to be exactly normal.
- C) The procedure is valid only if the two sample means are equal.

D) Sample sizes above 30 eliminate the need for independent observations.

APSTAT-MCQ4-036

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A test of a population proportion has power 0.80 against a particular alternative value. What is the probability of a Type II error at that alternative value?

- A) 0.80
- B) 0.20
- C) 0.05
- D) It cannot be found from the power.

APSTAT-MCQ4-037

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Two clinics report recovery rates by initial condition severity.

Clinic A: mild 18/20 recovered; severe 30/100 recovered

Clinic B: mild 80/100 recovered; severe 4/20 recovered

Clinic A has the higher recovery rate within both severity groups, yet Clinic B has the higher overall rate. Which explanation is best?

- A) Recovery status must have been measured quantitatively at Clinic B.
- B) The overall rates are invalid because each clinic treated 120 patients.
- C) Clinic A's higher subgroup rates prove its overall rate must also be higher.
- D) Severity is associated with both clinic composition and recovery, so combining groups with very different severity mixes reverses the comparison.

APSTAT-MCQ4-038

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Let $X \sim \text{Binomial}(100, 0.40)$. Using a normal approximation with continuity correction, what is $P(X \leq 35)$? The approximating normal distribution has mean 40 and standard deviation $\sqrt{24} \approx 4.899$. Use $\Phi(-0.92) = 0.1788$.

- A) 0.0668
- B) 0.1587
- C) 0.1788
- D) 0.8212

APSTAT-MCQ4-039

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A researcher planned a two-sided test comparing app-retention proportions. The observed difference favored the new app, and the two-sided p-value was 0.080. After seeing the data, the researcher reported a one-sided p-value of 0.040 and claimed significance at $\alpha = 0.05$. What is the main problem?

- A) The direction of the alternative was chosen after seeing the result, which invalidates the stated significance level.
- B) A one-sided p-value can never be smaller than a two-sided p-value.
- C) Retention proportions must always be analyzed with a t-test.
- D) Any p-value below 0.10 proves the new app has a large effect.

APSTAT-MCQ4-040

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A school tests a collaborative-learning app. Students in the same classroom are likely to share information about the app, creating spillover between treatment and control students. Which design best limits this interference?

- A) Assign individual students within each classroom independently to app or control and encourage discussion.
- B) Randomly assign entire classrooms to app or control, then account for classroom-level assignment in the analysis.
- C) Let teachers choose which students receive the app.

D) Give every classroom both conditions at the same time without tracking exposure.

APSTAT-MCQ4-041

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

A 95% confidence interval for a population mean is (4.1, 8.7). For a two-sided test using the same data at $\alpha = 0.05$, which null value would be rejected?

A) $H_0: \mu = 6.0$ only

B) $H_0: \mu = 4.1$ only

C) $H_0: \mu = 7.5$ only

D) $H_0: \mu = 0$ only

APSTAT-MCQ4-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Scores on a difficult final exam cluster near 88 points but have a long tail toward a few very low scores. Which relationship between the mean and median is most likely?

A) The mean is greater than the median because the distribution is left-skewed.

B) The mean equals the median because most scores are near 88.

C) The mean is less than the median because the low tail pulls the mean downward.

D) The relationship cannot be discussed for a quantitative variable.

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ4-001, Correct: B, Correct Answer: $z = 2.10$ and p-value 0.0357, Explanation: Reversing the order of the sample proportions multiplies the numerator, and therefore z , by -1 . A two-sided p-value depends on $|z|$, so it remains 0.0357.

APSTAT-MCQ4-002, Correct: D, Correct Answer: Weight group results according to the groups' shares of district enrollment., Explanation: Equal allocation gives small language groups more sample representation than their population shares. Population-share weights restore the proper contribution of each group to an overall district estimate.

APSTAT-MCQ4-003, Correct: A, Correct Answer: $(-4.74, -1.66)$ minutes, Explanation: $SE = 4.5/\sqrt{25} = 0.9$ minute and the margin of error is $1.711(0.9) = 1.540$ minutes. Thus the interval is $-3.2 \pm 1.540 =$

(-4.74, -1.66) minutes.

APSTAT-MCQ4-004, Correct: C, Correct Answer: The simulation estimates the probability of at least one missed connection as 0.274, with some Monte Carlo variability., Explanation: The simulated relative frequency is $2,740/10,000 = 0.274$. It estimates the long-run probability, but a finite simulation does not determine the exact probability or guarantee an exact future count.

APSTAT-MCQ4-005, Correct: A, Correct Answer: Restricting the range of age can weaken the observed correlation even if the broader linear tendency still exists., Explanation: Correlation depends on the variation represented in the data. Restricting age to a narrow interval can reduce the visible linear signal relative to other variability.

APSTAT-MCQ4-006, Correct: C, Correct Answer: Mobile has a 10-percentage-point higher first-contact resolution rate., Explanation: The mobile rate is $84/120 = 0.70$ and the desktop rate is $90/150 = 0.60$. The difference is 0.10, or 10 percentage points, in favor of mobile.

APSTAT-MCQ4-007, Correct: D, Correct Answer: Is the coached completion proportion higher than the uncoached proportion?, Explanation: $H_a: p_E - p_U > 0$ is equivalent to $p_E > p_U$, so it asks whether the coached population has the higher completion proportion.

APSTAT-MCQ4-008, Correct: B, Correct Answer: Candidate Q, because its SSE is smallest, Explanation: The least-squares line minimizes the sum of squared vertical residuals over all possible straight lines. Of the candidates shown, Q has the smallest SSE.

APSTAT-MCQ4-009, Correct: A, Correct Answer: Each residual is observed y – predicted y at the same x -value, so it is a vertical difference on the scatterplot., Explanation: Prediction fixes x and compares the observed response with the line's predicted response. Both are y -values, so the residual is measured vertically.

APSTAT-MCQ4-010, Correct: D, Correct Answer: The correlation remains -0.74 , but the new least-squares line is not generally the algebraic inverse of the original line., Explanation: Correlation is symmetric in x and y , but least-squares regression minimizes vertical residuals for a designated response. Reversing the roles changes that criterion, so the two fitted lines are not generally inverses.

APSTAT-MCQ4-011, Correct: C, Correct Answer: $7/12 \times 6/11 = 7/22$, Explanation: The first member is remote with probability $7/12$. Given that outcome, 6 of the 11 remaining members are remote, so $P(\text{both remote}) = (7/12)(6/11) = 7/22$.

APSTAT-MCQ4-012, Correct: B, Correct Answer: Sample thicknesses typically differ from the mean thickness by about 4.2 millimeters., Explanation: Standard deviation describes a typical distance of observations from their mean. It does not guarantee a fixed range or equal the IQR, and the 68% statement requires an approximately normal distribution.

APSTAT-MCQ4-013, Correct: D, Correct Answer: Because 80 exceeds 10% of 500, dependence from sampling without replacement may make the usual standard-error condition questionable., Explanation: The 10% condition supports treating observations as approximately independent when sampling without replacement. Here $80/500 = 0.16$, so the condition fails.

APSTAT-MCQ4-014, Correct: A, Correct Answer: Under H_0 , both samples are assumed to share one common population proportion, which the pooled proportion estimates., Explanation: The null equality justifies estimating one common p from the combined data for the test standard error. A confidence interval does not impose equality and therefore estimates each variance separately.

APSTAT-MCQ4-015, Correct: C, Correct Answer: The sample mean $\bar{x}$, centered at μ with standard deviation $\sigma/\sqrt{64}$, Explanation: The probability concerns an average across 64 days, so $\bar{x}$ is the relevant random variable. Its sampling distribution has mean μ and standard deviation $\sigma/\sqrt{n} = \sigma/\sqrt{64}$.

APSTAT-MCQ4-016, Correct: B, Correct Answer: 0.0668, Explanation: $z = (165 - 210)/30 = -1.50$. Therefore $P(X < 165) = P(Z < -1.50) = 0.0668$.

APSTAT-MCQ4-017, Correct: C, Correct Answer: Subject-to-subject variability can cancel in the differences, often reducing the standard error., Explanation: Pairing compares each subject with itself. When paired measurements are positively associated, between-subject variation is removed from the differences, often increasing precision.

APSTAT-MCQ4-018, Correct: D, Correct Answer: To assess possible response or classification error in the self-reported training status, Explanation: Comparing a random subset of self-reports with records provides evidence about misclassification or response bias. It does not change the original sampling method or variable type.

APSTAT-MCQ4-019, Correct: A, Correct Answer: Measure both variables on products sampled across the full intended density range, with adequate replication throughout that range., Explanation: Prediction across densities 20–80 requires paired density and absorption measurements representing that range. A single density, outcome-based selection, or estimated responses cannot establish the intended relationship.

APSTAT-MCQ4-020, Correct: B, Correct Answer: People connected to the initial participants may be overrepresented, so selection probabilities are unknown., Explanation: Recruitment follows social links from the initial participants, so people outside those networks can be missed and inclusion probabilities are not controlled.

APSTAT-MCQ4-021, Correct: D, Correct Answer: 0.26, Explanation: Exactly one valve succeeds in two disjoint ways: A succeeds and B fails, or A fails and B succeeds. The probability is $0.90(0.20) + 0.10(0.80) = 0.18 + 0.08 = 0.26$.

APSTAT-MCQ4-022, Correct: C, Correct Answer: Type I error probability decreases, but Type II error probability generally increases., Explanation: A lower α makes rejection harder, reducing false rejections under H_0 . With sample size and alternative fixed, that stricter threshold generally lowers power and raises the Type II error probability.

APSTAT-MCQ4-023, Correct: A, Correct Answer: 1.07, Explanation: The margin of error is half the interval width: $(18.6 - 14.2)/2 = 2.2$. Since $\text{margin} = t^* \times \text{SE}$, $\text{SE} = 2.2/2.064 \approx 1.07$.

APSTAT-MCQ4-024, Correct: D, Correct Answer: \$20–29, Explanation: With 40 orders, the median is determined by the 20th and 21st ordered values. The cumulative count through \$10–19 is $4 + 9 = 13$, and through \$20–29 it is 28, so both middle positions fall in the \$20–29 bin.

APSTAT-MCQ4-025, Correct: B, Correct Answer: $210/495 \approx 0.424$, Explanation: Choose 2 of the 5 damaged reels and 2 of the 7 undamaged reels. Thus $P(\text{exactly 2 damaged}) = C(5,2)C(7,2)/C(12,4) = 10(21)/495 = 210/495 \approx 0.424$.

APSTAT-MCQ4-026, Correct: B, Correct Answer: $460/495 \approx 0.929$, Explanation: Use the complement of selecting no damaged reels. $P(\text{at least 1 damaged}) = 1 - C(7,4)/C(12,4) = 1 - 35/495 = 460/495 \approx 0.929$.

APSTAT-MCQ4-027, Correct: C, Correct Answer: After the first reel is selected, the composition of the remaining reels changes, so the probability of damage on the second draw can change., Explanation: Without replacement, a damaged first reel leaves 4 damaged among 11, while an undamaged first reel leaves 5 damaged among 11. The second-draw probability depends on the first outcome.

APSTAT-MCQ4-028, Correct: A, Correct Answer: Sample 1 does not have at least 10 successes., Explanation: A two-proportion interval requires at least 10 successes and 10 failures in each sample. Sample 1 has only 8 successes; all other listed counts are at least 10.

APSTAT-MCQ4-029, Correct: D, Correct Answer: Observational unit: individual plant; variable: days from sowing until its first bloom, Explanation: The distribution consists of one quantitative time measurement for each plant. A day, greenhouse, or variety is not the unit on which this response is observed.

APSTAT-MCQ4-030, Correct: C, Correct Answer: 46, Explanation: The standard error is $8/\sqrt{16} = 2$. Since $-2.00 = (\bar{x} - 50)/2$, $\bar{x} - 50 = -4$ and $\bar{x} = 46$.

APSTAT-MCQ4-031, Correct: B, Correct Answer: Neighborhoods with higher mean book counts tend to have higher mean literacy scores, but the aggregate correlation does not determine the relationship for individual households., Explanation: The observational units are neighborhoods, so r describes association among neighborhood averages. It does not justify individual-level or causal claims.

APSTAT-MCQ4-032, Correct: A, Correct Answer: Parameter $p_M - p_C$ with $H_a: p_M - p_C > 0$, Explanation: The research question concerns population proportions, not sample statistics. "Raises" means the mentoring proportion exceeds the current-program proportion, so the ordered difference is positive.

APSTAT-MCQ4-033, Correct: C, Correct Answer: The replacements are not randomly selected and may systematically differ from nonresponding households., Explanation: Nearest responding households are convenience replacements, so the final sample no longer follows the original probability design and may not represent nonrespondents.

APSTAT-MCQ4-034, Correct: D, Correct Answer: Select one part at random, record its status, replace and remix it, and repeat 20 times., Explanation: Replacement keeps the refurbished probability at $12/50$ for every draw and supports independence. Without replacement, selecting 20 of only 50 parts changes the success probability substantially.

APSTAT-MCQ4-035, Correct: A, Correct Answer: The large sample sizes make the procedure reasonably robust to the moderate skewness., Explanation: Independent samples of 60 and 55 with no outliers are large enough for two-sample t methods to tolerate moderate skewness. Large size does not replace the independence requirement.

APSTAT-MCQ4-036, Correct: B, Correct Answer: 0.20, Explanation: Power = $1 - P(\text{Type II error})$ at the specified alternative. Therefore $\beta = 1 - 0.80 = 0.20$.

APSTAT-MCQ4-037, Correct: D, Correct Answer: Severity is associated with both clinic composition and recovery, so combining groups with very different severity mixes reverses the comparison., Explanation: Clinic A recovers 90% of mild and 30% of severe patients, exceeding Clinic B's 80% and 20%. But Clinic A treats far more severe cases, so its overall rate is $48/120 = 40\%$ versus $84/120 = 70\%$ for B. Aggregation reverses the subgroup comparison.

APSTAT-MCQ4-038, Correct: C, Correct Answer: 0.1788, Explanation: Apply the continuity correction: $P(X \leq 35)$ is approximated by $P(Y \leq 35.5)$. Then $z = (35.5 - 40)/4.899 \approx -0.92$, so the probability is approximately 0.1788.

APSTAT-MCQ4-039, Correct: A, Correct Answer: The direction of the alternative was chosen after seeing the result, which invalidates the stated significance level., Explanation: A one-sided alternative must be specified before examining the direction of the data. Selecting the favorable tail afterward inflates the chance of a false-positive claim.

APSTAT-MCQ4-040, Correct: B, Correct Answer: Randomly assign entire classrooms to app or control, then account for classroom-level assignment in the analysis., Explanation: Assigning whole classrooms limits exposure across conditions within a classroom. Because assignment occurs by classroom, the analysis must also recognize classroom clustering.

APSTAT-MCQ4-041, Correct: D, Correct Answer: $H_0: \mu = 0$ only, Explanation: A two-sided $\alpha = 0.05$ test rejects null values outside the corresponding 95% confidence interval. Values 4.1, 6.0, and 7.5 lie in the interval, while 0 does not.

APSTAT-MCQ4-042, Correct: C, Correct Answer: The mean is less than the median because the low tail pulls the mean downward., Explanation: A long lower tail makes the distribution left-skewed. The mean is pulled toward that tail more strongly than the median.

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ4

· APSTAT-MCQ4-001 to 005: Unit 3 reverses the group order in a two-proportion z-test, changing the sign of z while preserving its two-sided p-value; Unit 1 equal-allocation language-group sampling requires population-share weights for an overall district estimate; Unit 4 paired employee task-time differences construct a 90% t-interval from $\bar{x}_d$, s_d , n, and t^* ; Unit 2 transit-schedule simulation interprets 2,740 missed-connection days in 10,000 trials as a Monte Carlo estimate with variability; Unit 5 age and reaction-speed correlations illustrate attenuation from restricting the explanatory-variable range. Misconceptions include changing the two-sided p-value when the order reverses, treating disproportionate strata equally, using s_d instead of its standard error, treating a simulated relative frequency as an exact probability or guaranteed future count, and assuming subgroup restriction cannot change r.

· APSTAT-MCQ4-006 to 010: Unit 1 help-desk access-method counts compare conditional resolution rates by percentage-point difference; Unit 3 coaching completion translates $H_a: p_E - p_U > 0$ into its research question; Unit 5 three-question archaeology stimulus selects the least-squares candidate by minimum SSE, defines vertical y-residuals, and explains that swapping x and y preserves r but does not generally invert the regression line. Misconceptions include comparing raw resolved counts, reversing an ordered parameter difference, selecting the largest SSE, measuring horizontal or perpendicular

residuals, and treating the two regression lines as algebraic inverses.

- APSTAT-MCQ4-011 to 015: Unit 2 selects two project-team members without replacement and multiplies conditional probabilities; Unit 1 ice thickness interprets standard deviation as typical distance from the mean without assuming normality; Unit 4 account-balance SRS samples 80 of 500 and detects failure of the 10% condition; Unit 3 explains pooled standard error under $H_0: p_1 = p_2$ versus unpooled estimation; Unit 2 bakery daily revenue identifies the sample-mean distribution with center μ and spread $\sigma/\sqrt{64}$. Misconceptions include treating draws as independent, equating standard deviation with IQR or a guaranteed range, ignoring finite-population dependence, pooling merely because sample sizes differ, and using a total or proportion for an average-revenue probability.

- APSTAT-MCQ4-016 to 020: Unit 1 approximately normal household electricity use finds a lower-tail area from $z = -1.50$; Unit 4 positively associated repeated measurements explain why pairing can reduce standard error; Unit 3 training survey validates self-reports against confidential records to assess classification error; Unit 5 material-density sampling covers the intended prediction range with paired measurements; Unit 1 snowball sampling of a hard-to-identify financial-habits population exposes unknown selection probabilities and network overrepresentation. Misconceptions include taking the complementary normal tail, assuming pairing forces zero mean, treating validation as random sampling, collecting data at one x -value, and labeling snowball recruitment as probability sampling.

- APSTAT-MCQ4-021 to 025: Unit 2 independent irrigation valves calculate exactly one success by adding two disjoint orders; Unit 3 lowering α trades lower Type I error for higher Type II error at fixed n and alternative; Unit 4 recovers a standard error from confidence-interval width and t^* ; Unit 1 unequal order-value histogram bins locate the median class from cumulative frequencies; Unit 2 archive film-reel set begins with a hypergeometric exactly-two-damaged probability using combinations. Misconceptions include counting only one success order, assuming both error types fall with α , confusing margin with standard error, selecting the tallest bin without locating middle positions, and omitting either damaged or undamaged combinations.

- APSTAT-MCQ4-026 to 030: Unit 2 film-reel set uses a no-damage complement and explains dependence under sampling without replacement; Unit 3 two-proportion interval counts detect only 8 successes in sample 1; Unit 1 first-bloom research identifies individual plants and quantitative days-to-bloom; Unit 4 reverses the one-sample t formula to recover $\bar{x}$ from μ_0 , s , n , and t . Misconceptions include subtracting the wrong event, keeping second-draw probability fixed, checking pooled null counts for an interval, confusing observational unit with variable, and omitting $\sqrt{n}$ when solving backward.

- APSTAT-MCQ4-031 to 035: Unit 5 neighborhood-average books and literacy correlation warns against individual-level and causal conclusions from aggregated data; Unit 3 mentoring completion defines an ordered population-proportion difference and right-tailed alternative; Unit 1 mail-survey nearest-neighbor replacement breaks probability sampling after nonresponse; Unit 2 refurbished parts use replacement and remixing to create 20 constant-probability independent trials; Unit 4 large independent samples with moderate skew and no outliers justify robust two-sample t inference. Misconceptions include ecological inference, using $\hat{p}$ in a parameter, treating convenience replacements as random, ignoring changing probabilities in a large sampling fraction, and believing large samples remove independence requirements.

- APSTAT-MCQ4-036 to 040: Unit 3 converts power 0.80 to Type II probability $\beta = 0.20$; Unit 1 clinic recovery tables exhibit an aggregate reversal from different severity mixes despite higher Clinic A rates in each stratum; Unit 2 applies a continuity-corrected normal approximation to $P[\text{Binomial}(100, 0.40) \leq 35]$; Unit 3 rejects post-hoc conversion of a two-sided retention test to a favorable one-sided test; Unit 1 classroom-level random assignment limits spillover in a collaborative-app experiment and requires cluster-aware analysis. Misconceptions include equating power with β , ignoring a lurking stratification variable, omitting the 35.5 continuity correction, choosing direction after data inspection, and randomizing individuals when interference is likely.

- APSTAT-MCQ4-041 to 042: Unit 4 uses equivalence between a 95% mean interval and a two-sided $\alpha = 0.05$ test to identify only $\mu_0 = 0$ as rejected; Unit 1 left-skewed final-exam scores infer mean below median because of the low tail. Misconceptions include rejecting null values inside the interval and reversing the direction in which skew pulls the mean.
- Unit counts for APSTAT-MCQ4: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: archaeology soil-depth/artifact-density SSE regression comparison, APSTAT-MCQ4-008 to 010; archive nitrate-damaged film-reel sampling without replacement, APSTAT-MCQ4-025 to 027.